

Predictive Maintenance Using IoT and AI

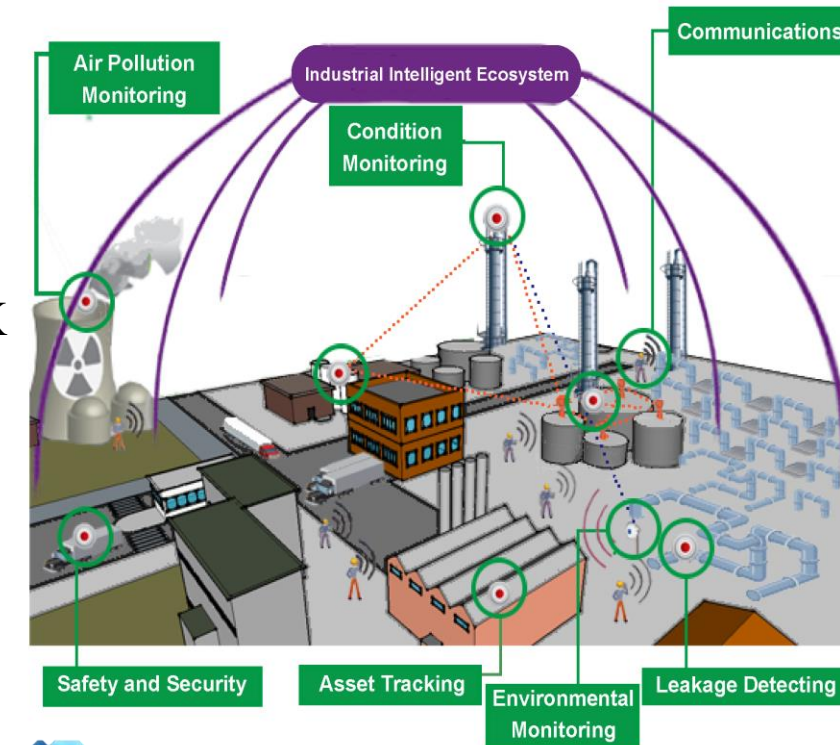
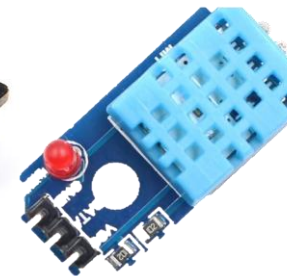
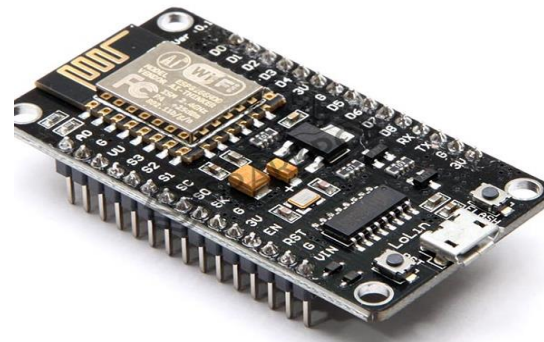
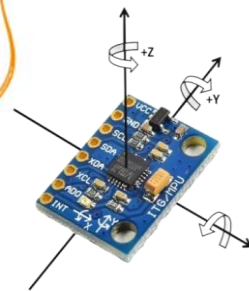
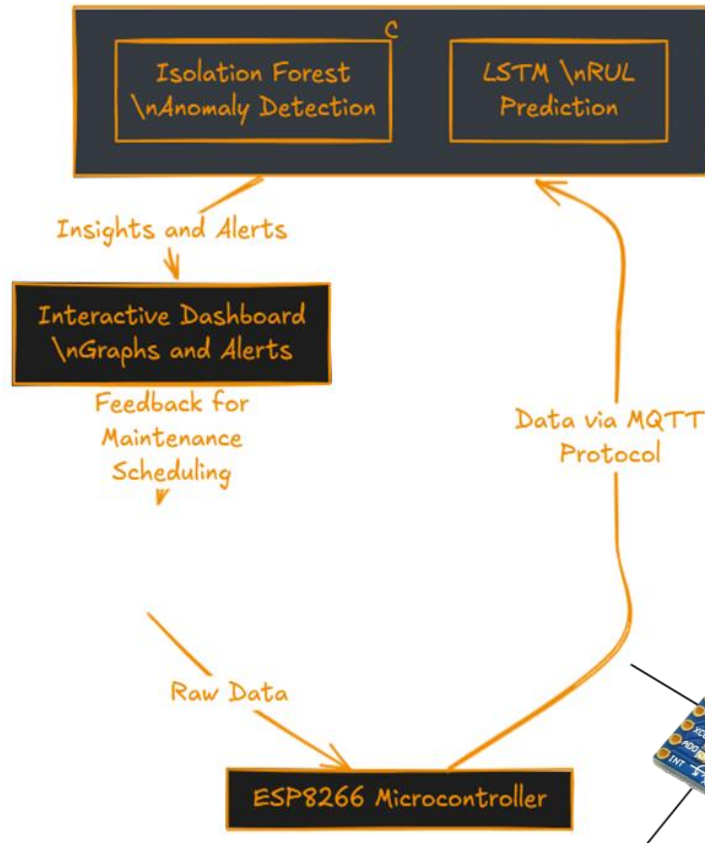
for
Seminar Presentation

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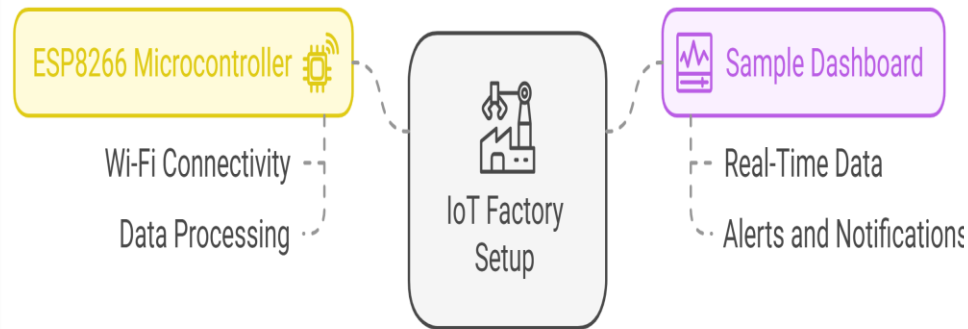
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Introduction

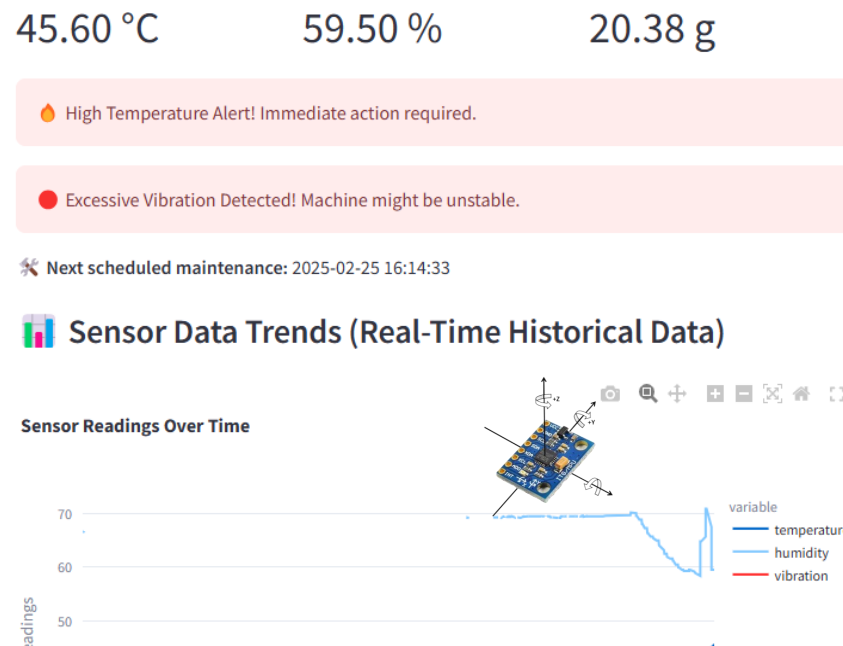


Predictive maintenance has emerged as a transformative solution for modern industries, addressing the limitations of traditional, reactive maintenance strategies. By leveraging IoT sensors and AI-driven algorithms, it enables real-time monitoring of equipment, anomaly detection, and failure prediction. This project integrates cutting-edge technologies such as ESP8266 microcontrollers, MQTT protocol, and machine learning algorithms (Isolation Forest and LSTM) to develop a cost-effective and efficient predictive maintenance system. The approach reduces unplanned downtime, optimizes maintenance schedules, and enhances operational efficiency in industrial settings.

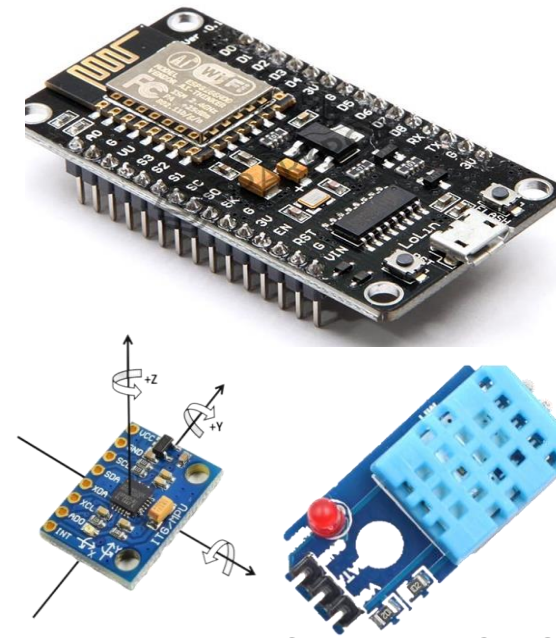
IoT Factory Setup: Components and Functionality



Fig_No:1 Basic Flow PdM Using IOT



Fig_No:2 Dashboard

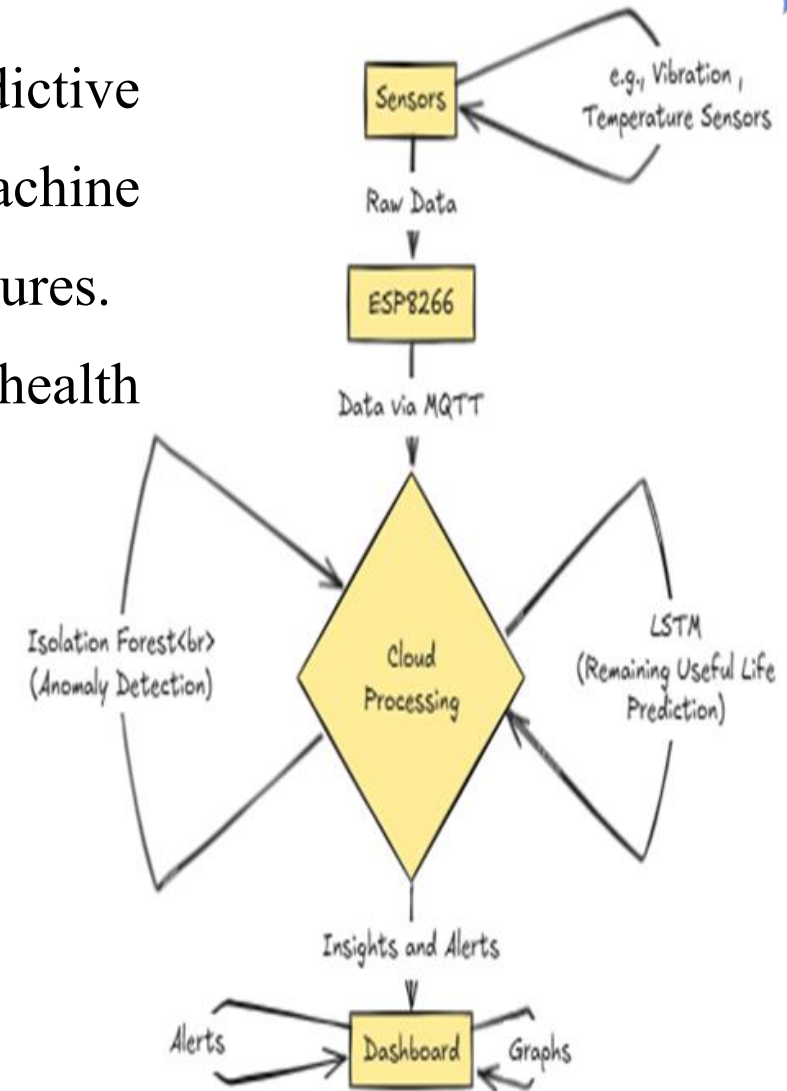


Fig_No:3 IOT DEVICES

Objective



- The objective of this project is to develop an IoT-based predictive maintenance system that leverages real-time sensor data and machine learning algorithms to detect anomalies and predict equipment failures.
- Develop an IoT-based system for real-time equipment health monitoring.
- Use ESP8266 with temperature & vibration sensors to collect machine data.
- Implement AI models like:
 - Isolation Forest for anomaly detection.
 - LSTM (Long Short-Term Memory) for Remaining Useful Life (RUL) prediction.

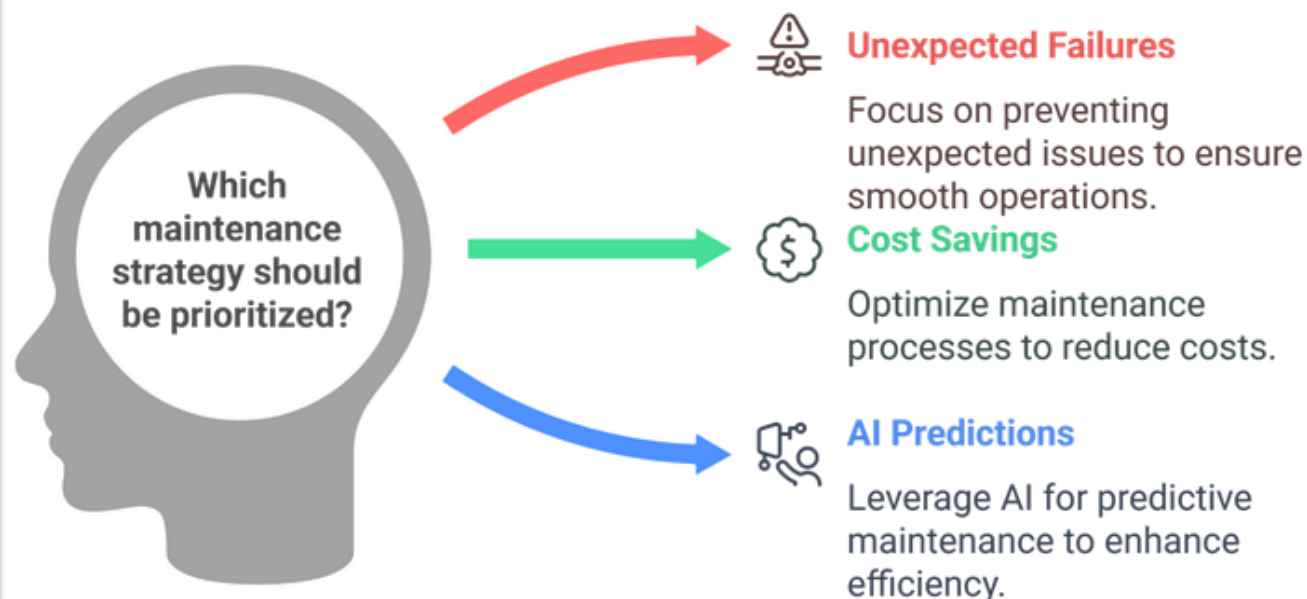


Fig_No:4 FlowChart

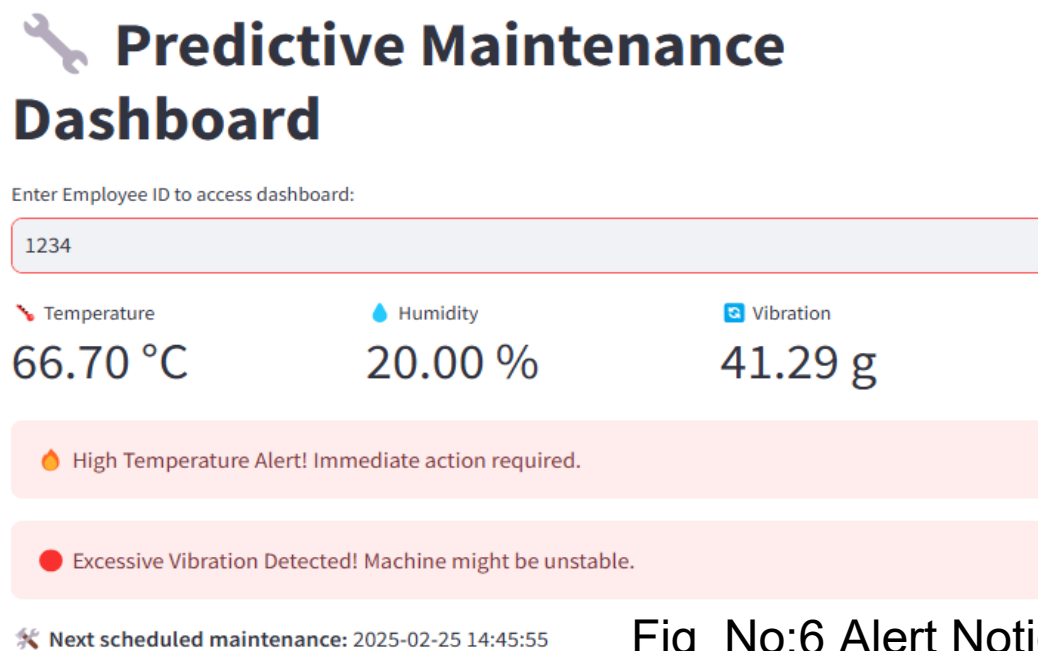
Problem Statement



- Industries face significant challenges due to **unexpected equipment failures**, leading to unplanned downtime, high maintenance costs, and productivity losses.
- Traditional maintenance approaches—**reactive (fix after failure)** and **preventive (scheduled repairs)**—are inefficient and costly.
- There is a pressing need for a smart, data-driven system that can **predict failures in advance** that can analyze real-time sensor data and detect anomalies early to reduce unexpected breakdowns.



Fig_No:5



Fig_No:6 Alert Notice

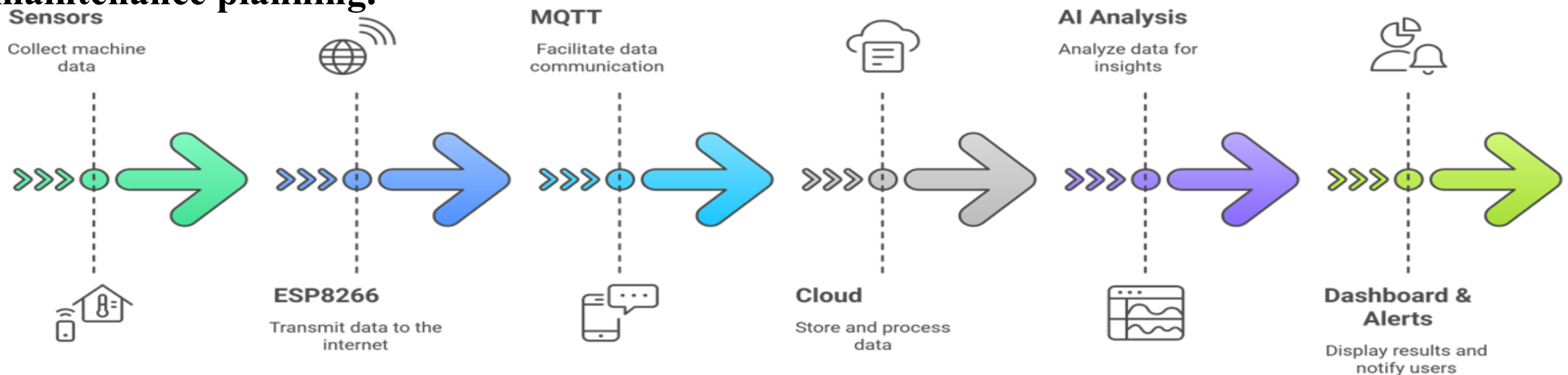
Methodology



The Predictive Maintenance System follows a structured approach that integrates:

- 1** IoT Device & Sensors(ESP8266, DHT11, MPU6050):-Captures live temperature, humidity & vibration data.
- 2** Data Transmission (MQTT Protocol):- Securely sends sensor readings to the cloud/database.
- 3** Data Storage (SQLite Database) :-Stores timestamped sensor values for analysis.
- 4** Machine Learning Models (Isolation Forest & LSTM):-Detect anomalies and predict remaining useful life.
- 5** Dashboard & Alerts (Streamlit) :-Displays real-time sensor data, alerts users of anomalies, and schedules maintenance.

This methodology ensures **real-time monitoring, early failure detection, and optimized maintenance planning.**



Fig_No:7 PdM Data flow

Working Architecture/Model



Model integrates IoT & AI to monitor equipment health in real-time. The Predictive Maintenance System follows a structured workflow:

❶ Sensor Data Collection & Transmission

- ESP8266 reads real-time sensor data from DHT11 (Temperature, Humidity) & MPU6050 (Vibration).
- Data is sent via MQTT protocol to the cloud broker (HiveMQ).

❷ Data Storage & Processing (SQLite Database)

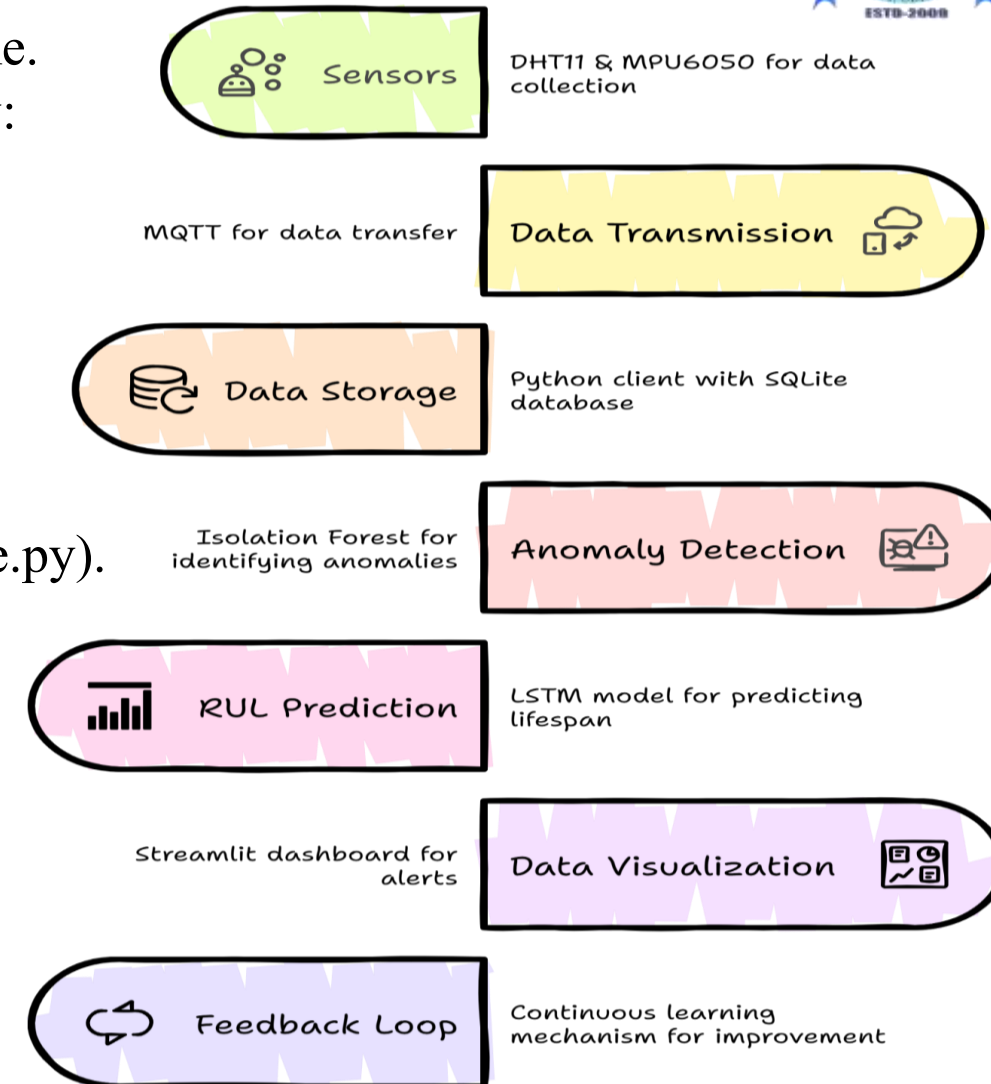
- MQTT broker forwards sensor data to a Python script (mqtt2sqlite.py).
- Sensor readings are stored in an SQLite database for analysis.

❸ Anomaly Detection & RUL Prediction

- Isolation Forest Model detects anomalies in real-time.
- LSTM Model predicts Remaining Useful Life (RUL) using past sensor trends.

❹ Dashboard & Alerts (Streamlit)

- Live visualization of real-time and historical sensor data.
- Triggers alerts when an anomaly or low RUL is detected.



Fig_No:8

Algorithm (pseudo code)



Anomaly Detection - Isolation Forest

Input: Real-time Sensor Data (Temperature, Vibration, Humidity)

Output: Anomaly Classification (-1: Anomaly, 1: Normal)

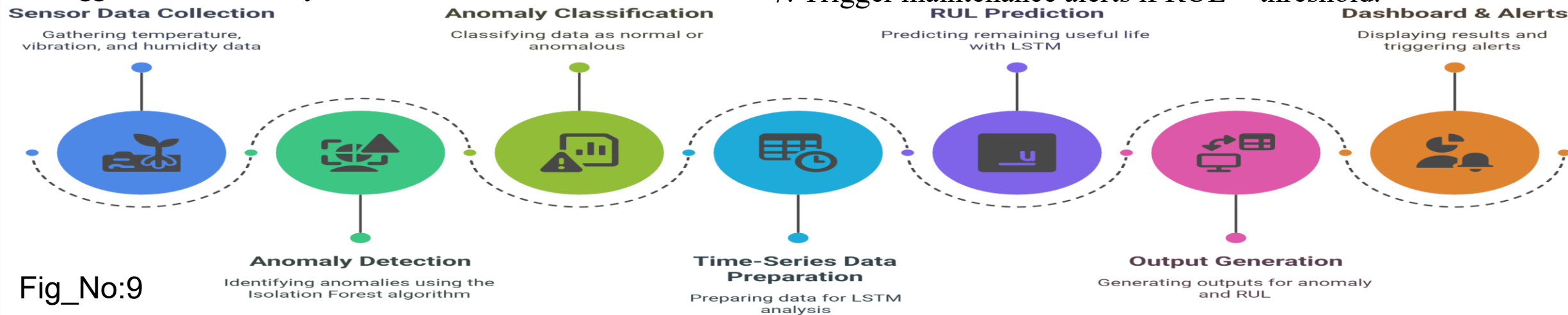
1. Fetch the latest sensor data from SQLite database.
2. Preprocess data (normalize, handle missing values).
3. Load pre-trained Isolation Forest model.
4. Predict anomaly score for incoming data:
 - If anomaly score $>$ threshold $\rightarrow$ classify as anomaly (-1).
 - Else $\rightarrow$ classify as normal (1).
5. Store anomaly classification in SQLite database.
6. Trigger alerts if anomaly detected.

Remaining Useful Life (RUL) Prediction - LSTM

Input: Latest time-series sensor data (Temperature, Vibration, Humidity)

Output: Predicted Remaining Useful Life (Days)

1. Fetch the latest time-series data window from SQLite.
2. Preprocess data (normalize, sequence formatting for LSTM).
3. Load pre-trained LSTM model.
4. Predict RUL based on recent sensor trends.
5. Store predicted RUL values in database.
6. Update dashboard with RUL forecast.
7. Trigger maintenance alerts if $RUL <$ threshold.



Fig_No:9

Implementation



The **Predictive Maintenance System** is implemented using **IoT, AI, and real-time analytics**:

1 Hardware Components (ESP8266 + Sensors)

- ESP8266 collects real-time sensor data from DHT11 (Temperature, Humidity) & MPU6050 (Vibration).
- Sends data wirelessly via MQTT to a cloud broker.

2 Data Storage & Processing (SQLite + MQTT Broker)

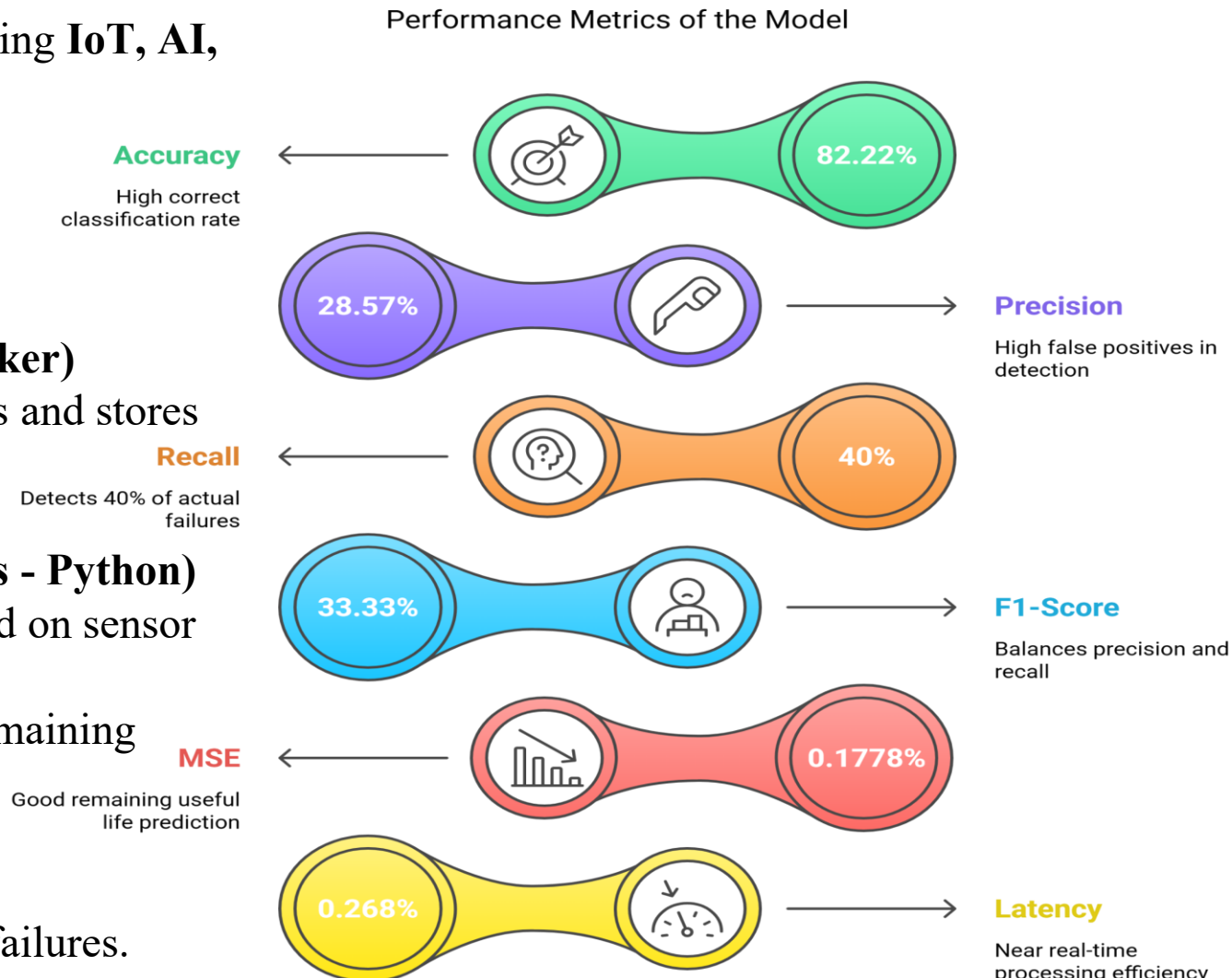
- Python script (mqtt2sqlite.py) captures MQTT messages and stores them in SQLite database.
- Data is structured for machine learning analysis.

3 Anomaly Detection & RUL Prediction (AI Models - Python)

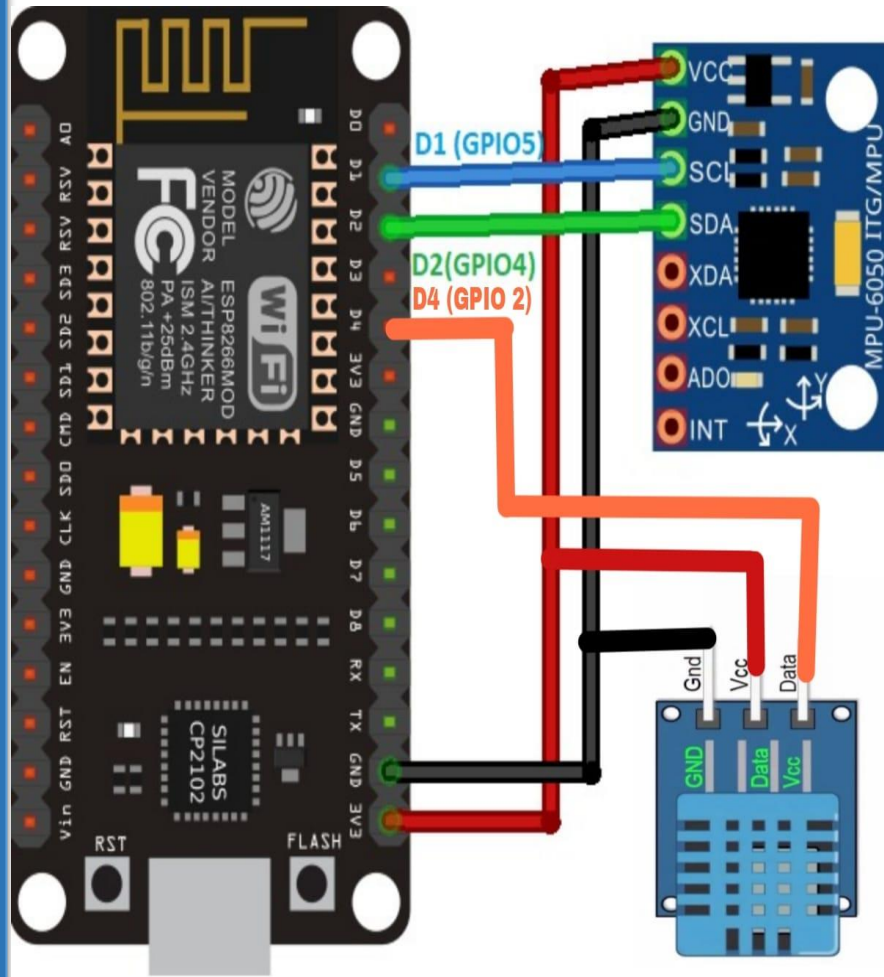
- Isolation Forest Model detects real-time anomalies based on sensor values.
- LSTM Model analyzes time-series data and predicts Remaining Useful Life (RUL).

4 Visualization & Alerts (Streamlit Dashboard)

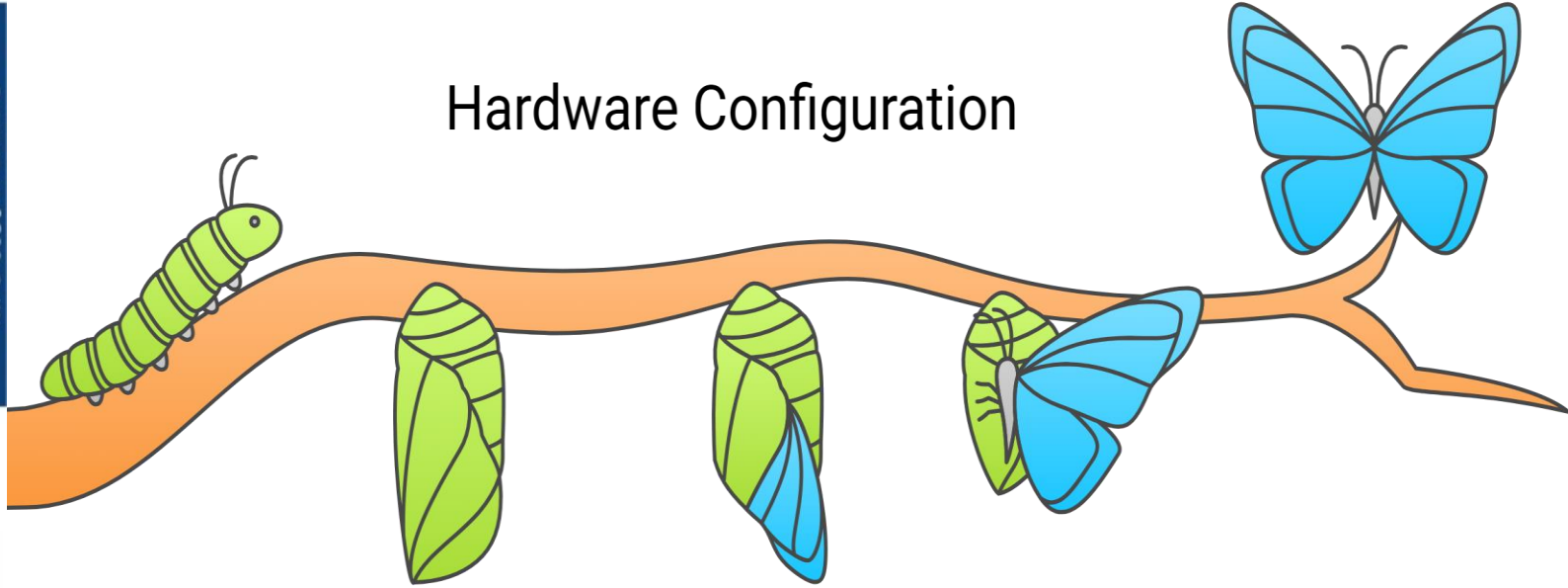
- Real-time monitoring using interactive graphs.
- Automatic alerts triggered for anomalies and predicted failures.



Fig_No:10 Performance Metrics Of The Model



Hardware Configuration



Initial Setup

Basic sensor data collection
DHT11 → Captures temperature & humidity.
MPU6050 → Measures machine vibration levels.

Integrate Sensors

Connect DHT11 and MPU6050

Enable Wi-Fi

Activate ESP8266 Wi-Fi module

Establish Protocol

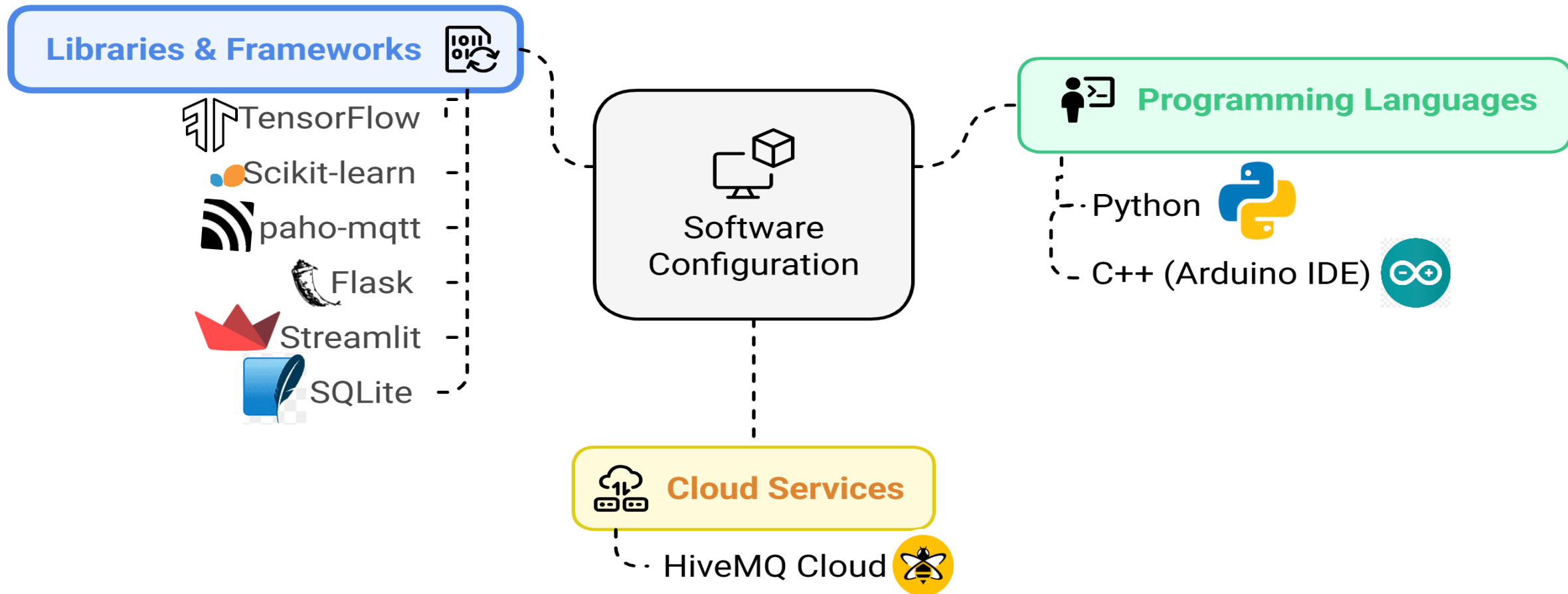
Implement MQTT for data transmission

Cloud Connectivity

Real-time data in the cloud

Fig_No:11 Hardware Configuration Flow

Software Configuration



Fig_No:12 Software Configuration

Results



Predictive Maintenance Dashboard

Enter Employee ID to access dashboard:

321

Temperature

45.60 °C

Humidity

59.50 %

Vibration

20.38 g

High Temperature Alert! Immediate action required.

Excessive Vibration Detected! Machine might be unstable.

Next scheduled maintenance: 2025-02-25 16:14:33

Fig_No:13

An employee authentication.

Live Sensor Readings.

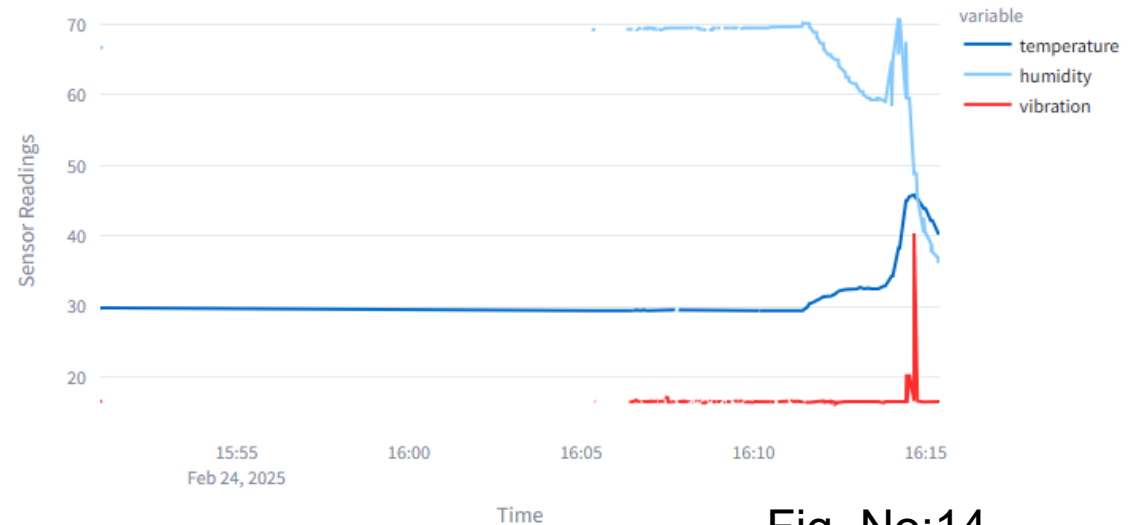
- Temperature: 45.60°C (High Alert 🚨)
- Humidity: 59.50%
- Vibration: 20.38g (Excessive Alert 🚨)

Critical Alerts.

- High Temperature Alert: Immediate action required.
- Excessive Vibration Detected: Machine might be unstable.
- Next Scheduled Maintenance: 2025-02-25 16:14:33

Sensor Data Trends (Real-Time Historical Data)

Sensor Readings Over Time



Fig_No:14

- **Temperature (Dark Blue Line)**: Shows a stable trend initially but experiences a sharp rise and fall, indicating a possible anomaly.
- **Humidity (Blue Line)**: Follows a consistent pattern with minor fluctuations.
- **Vibration (Red Line)**: Sudden spike indicates an anomaly that might be a fault or mechanical failure.

Observation



❶ Critical Alerts Indicate Potential Machine Failure 🔔

- Temperature exceeded safe limits (45.60°C) → High risk of overheating, potential component damage.
- Excessive vibration detected (20.38g) → Suggests mechanical imbalance, misalignment, or wear & tear.
- Immediate intervention required to prevent unexpected breakdowns.

❷ Predictive Maintenance Scheduling is Effective 🛠️

- System automatically scheduled maintenance for 2025-02-25 16:14:33.
- Reduces unplanned downtime, ensuring continuous operations.
- Provides real-time monitoring to trigger early alerts & corrective actions.

❸ Sensor Data Trends Reveal Anomalies 📊

- Temperature & Vibration showed a sudden increase → Suggests a possible malfunction or overload condition.
- Humidity levels are within safe limits (59.50%), so no immediate concern for moisture-related failures.

❹ System Security & Accessibility 🔒

- Employee authentication (ID: 321) required for dashboard access → Prevents unauthorized modifications.
- Only authorized personnel can monitor and respond to alerts, enhancing data integrity

Application



1 Industrial Manufacturing

- Detects machine wear & tear early → Avoids unexpected downtime & costly failures.
- Enhances production efficiency by reducing maintenance costs.

2 Power Plants & Energy Sector

- Monitors critical equipment (e.g., turbines, transformers) to prevent overheating & failures.
- Ensures stable power supply by predicting faulty components.

3 Automotive & Heavy Machinery

- Detects excessive vibration & temperature spikes in engines & heavy-duty machines.
- Improves vehicle safety by alerting before breakdowns.

4 Aerospace & Aviation

- Ensures aircraft safety by monitoring engine temperature & structural vibrations.
- Reduces maintenance time by using real-time predictive analytics.

5 Smart Buildings & IoT-Enabled Infrastructure

- Monitors HVAC systems to prevent overheating & inefficiency.
- Enhances smart city infrastructure by predicting electrical & mechanical failures.

Advantage and Disadvantage



Advantage

1 Reduced Downtime

- Prevents unexpected failures, keeping machines running without interruptions.

2 Cost Savings

- Reduces maintenance costs by fixing issues before breakdowns.
- Saves on labor, repair, & part replacement expenses.

3 Increased Equipment Lifespan

- Early fault detection minimizes wear & tear, extending machine life.

4 Improved Safety & Reliability

- Detects dangerous conditions (e.g., overheating, excessive vibration) to prevent accidents.

5 Data-Driven Decision Making

- Uses AI & IoT sensors for real-time monitoring, leading to better planning & resource allocation.

Disadvantage

1 High Initial Cost

- Expensive to implement due to sensor installation, AI models, & hardware setup.

2 Requires Skilled Workforce

- Needs data analysts & AI experts to handle predictive algorithms & sensor management.

3 Data Dependency & False Alerts

- Errors in sensor data can cause false maintenance alerts or miss real issues.

4 Integration Challenges

- Complex integration with existing legacy systems in industries.

5 Security & Privacy Risks

- IoT sensors are vulnerable to cyber threats & data breaches if not secured properly.

Conclusion & Future Scope



Conclusion:

The Smart Predictive Maintenance System successfully detects anomalies in temperature, humidity, and vibration using IoT sensors and AI. It helps prevent equipment failures, reducing downtime and maintenance costs while enhancing safety and efficiency in industries.

Future Scope:

- AI-Driven Analytics: Advanced machine learning for precise failure prediction.
- Cloud-Based Monitoring: Remote access through cloud platforms for global monitoring.
- Automated Maintenance: AI-driven scheduling to reduce manual intervention.
- Edge Computing: Faster real-time processing without cloud dependency.
- Cybersecurity Enhancements: Blockchain & encryption for secure IoT data.

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1. *Industrial IoT Data Collection & Analysis for Real-Time Decision-Making and Predictive Maintenance*, RTInsights. [Link for E-book](#) .

Thank You

“With predictive maintenance  powered by IoT 
and AI , we’re not just fixing problems  —
we’re preventing them  before they happen  .”